

POC: AI-Assisted Document Creation Service

1. Project Overview

This Proof of Concept (POC) describes the **AI-Assisted Document Creation Service**.

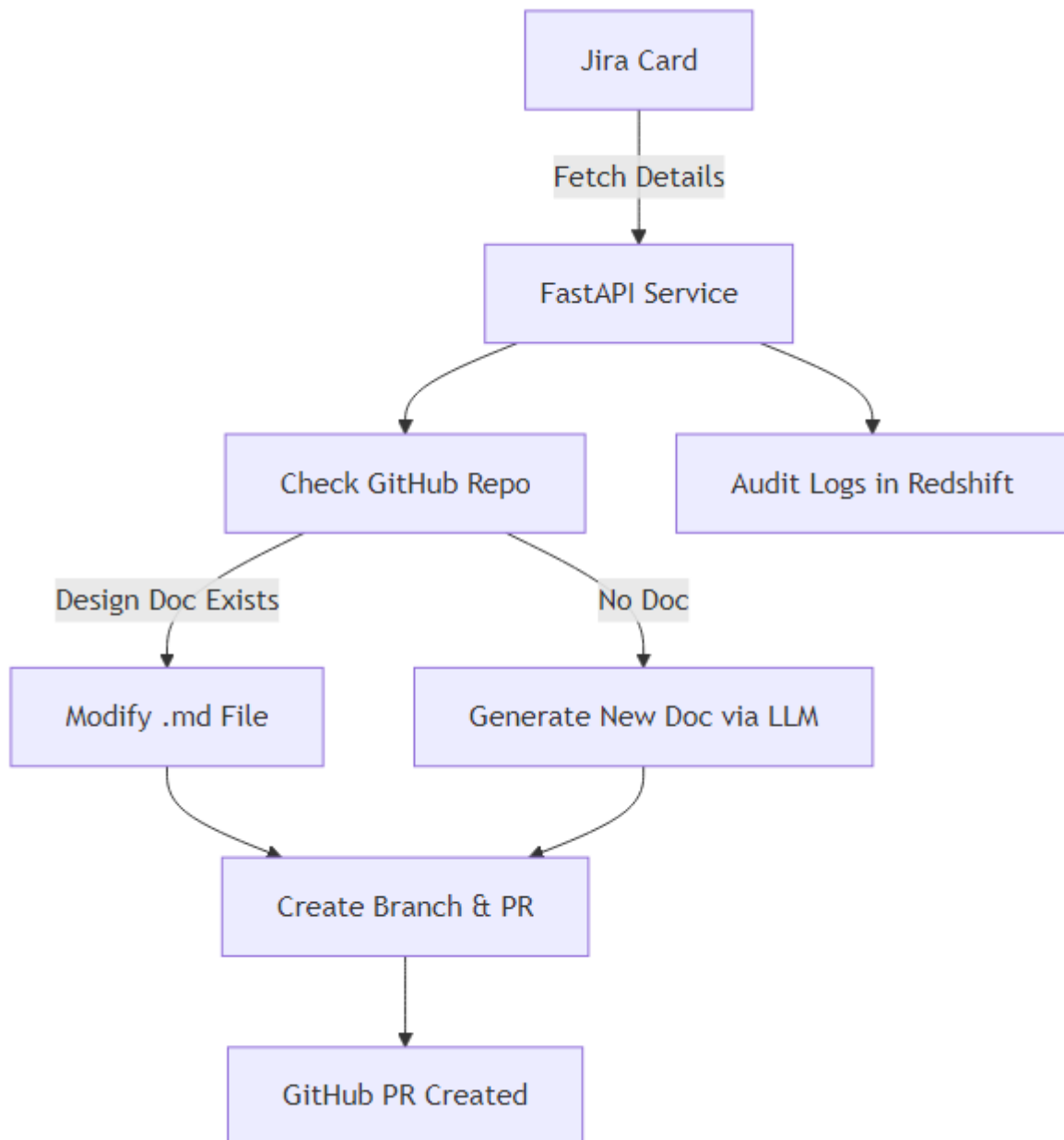
The system integrates **Jira, GitHub, OpenAI (ChatGPT), and AWS Redshift** to automate the creation and management of design documents. It supports both **UI-driven** and **API-driven** workflows.

2. Detailed Requirements

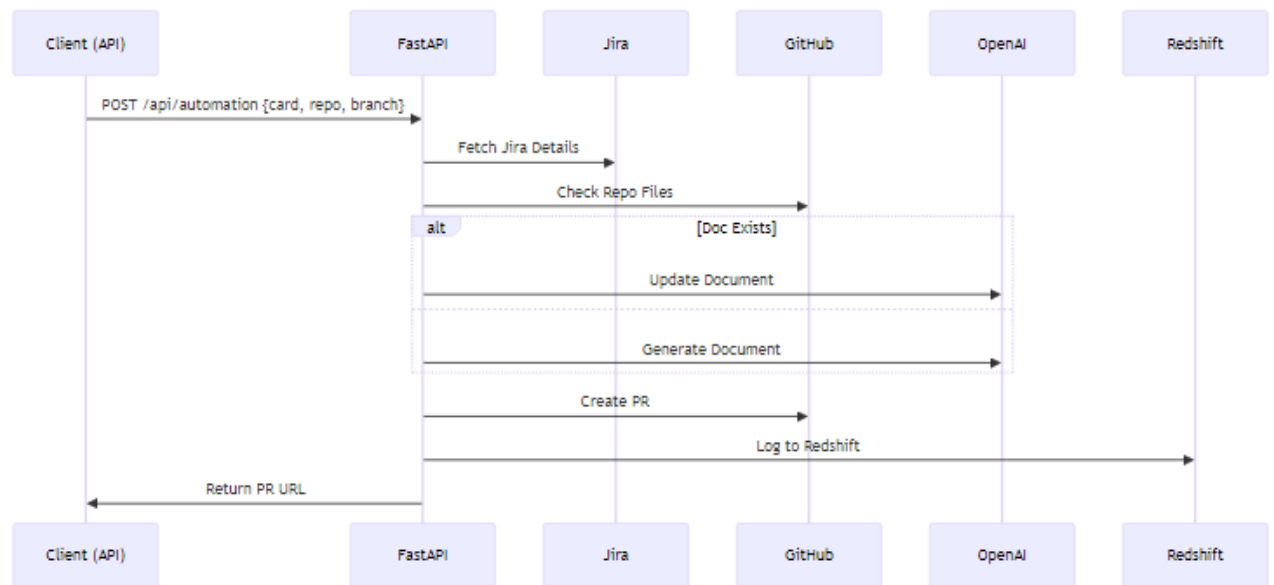
1. Jira integration to fetch card details.
 2. GitHub integration to scan repository files.
 3. If design docs exist → update them, else generate new ones.
 4. Create feature branch and raise PR automatically.
 5. Logs stored in AWS Redshift.
 6. Supports UI via Jinja templates and API calls.
 7. Deployed in Docker container on AWS ECS/App Runner.
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3. High-Level Architecture

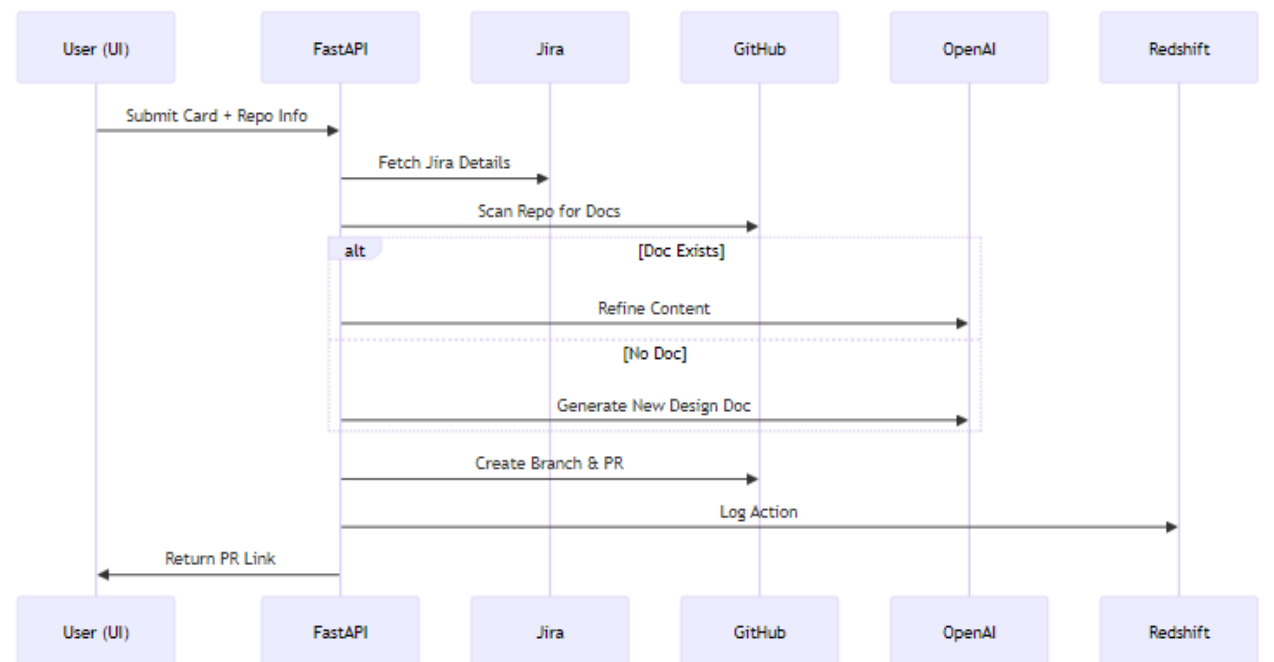
The system is designed with **FastAPI** at its core, serving both UI and API endpoints. It uses **AWS Secrets Manager** for credentials, **Redshift** for auditing, and **Dockerized deployments** in AWS ECS/App Runner.



4. Sequence Diagram – UI Flow



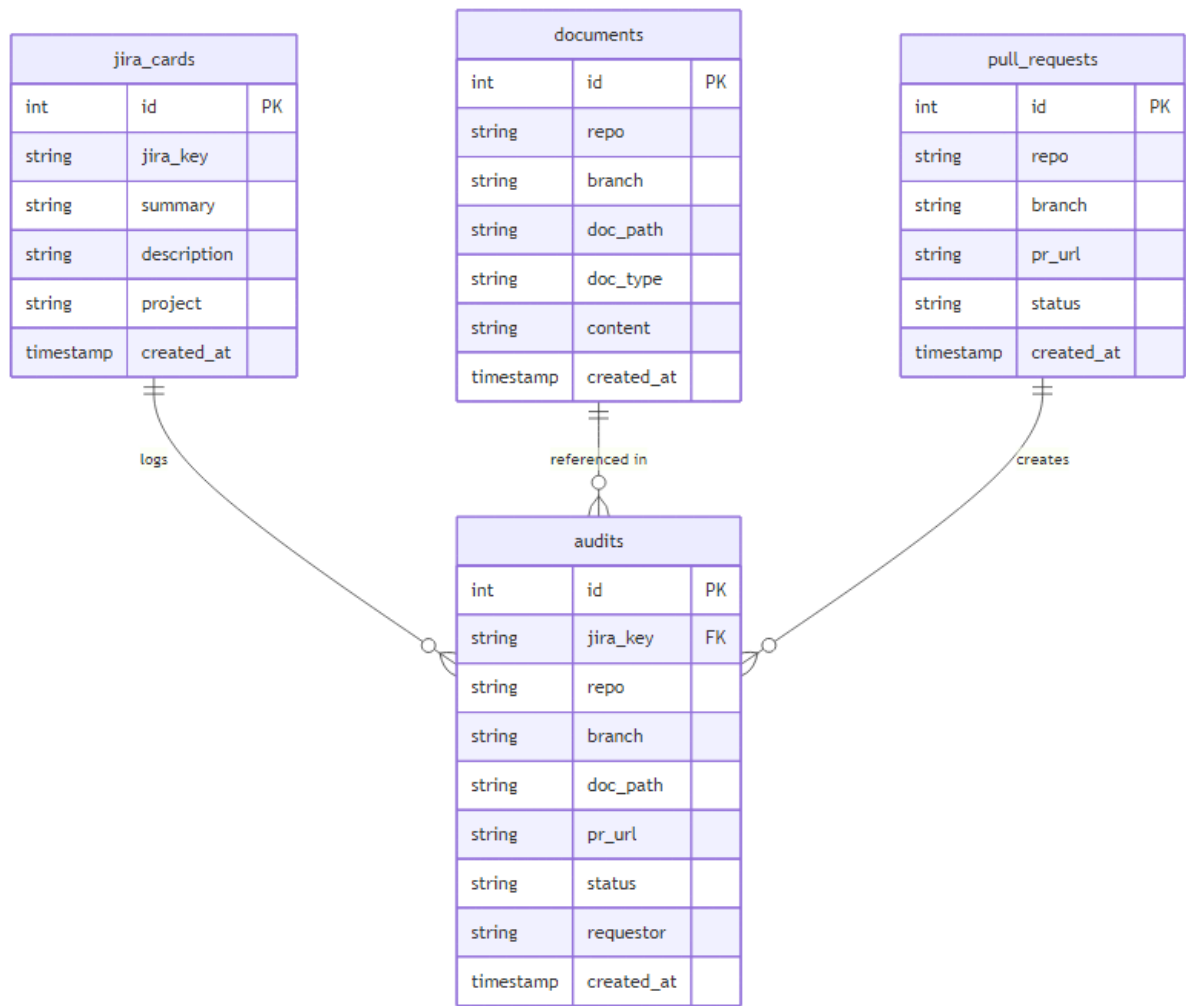
5. Sequence Diagram – API Flow



6. Database Schema & ERD

The database schema uses **AWS Redshift** for auditing and analytics.

Tables: **audits**, **jira_cards**, **documents**, **pull_requests**



7. API Contracts

POST /api/automation

Request:

```
{
  "card": "JIRA-123",
  "project": "project-x",
  "repo": "repo-name",
  "branch": "feature-abc",
  "doc_path": "docs/design.md"
}
```

Response:

```
json

{
  "status": "success",
  "pr_url": "https://github.com/org/repo/pull/123"
}
```

GET /audits

Response:

```
[
  {
    "jira_key": "JIRA-123",
    "repo": "repo-name",
    "branch": "feature-abc",
    "doc_path": "docs/design.md",
    "pr_url": "https://github.com/org/repo/pull/123",
    "status": "created",
    "created_at": "2025-09-03T12:00:00Z"
  }
]
```

8. CI/CD Pipeline

Using **GitHub Actions**:

1. Run lint/tests.
 2. Build Docker image.
 3. Push to ECR.
 4. Deploy to ECS/App Runner.
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9. Security & Rollback

- Store credentials in **AWS Secrets Manager**.
 - IAM roles restrict service access.
 - Rollback by redeploying previous Docker image.
 - ECS supports **blue/green deployments**.
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10. End-to-End User Flow

UI User

1. Open webapp.
2. Submit Jira ID + repo info.
3. Receive PR link.

API Consumer

1. Call POST /api/automation.
 2. Receive PR link.
 3. Optionally track until merge.
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